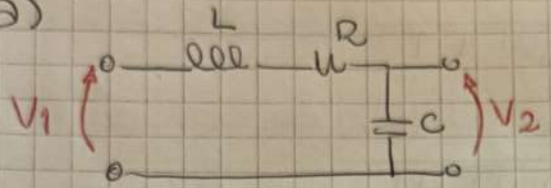
TS 6

a)



$$H(s) = \frac{V_2(s)}{V_1(s)}$$

$$H(s) = \frac{1/sC}{sL + R + 1/sC}$$

FORMA CANONICA

$$s^2 + \frac{\omega_0}{Q} s + \omega_0^2$$

$$H(s) = \frac{1}{s^2 LC + sRC + 1}$$

$$H(s) = \frac{1/LC}{s^2 \frac{LC}{LC} + s\frac{RC}{LC} + \frac{1}{LC}}$$

$$H(s) = \frac{1/LC}{s^2 + \frac{R}{L} s + 1/LC}$$

$$\omega_0^2 = \frac{1}{LC} \rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow LC = 1$$

$$\frac{\omega_0}{Q} = \frac{R}{L} \rightarrow \frac{R}{L} = \frac{1}{\frac{\sqrt{2}}{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\boxed{H(s) = \frac{1}{s^2 + \sqrt{2}\, s + 1}}$$

EVALUANDO EN $S = j\omega$

$$H(j\omega) = \frac{1}{-\omega^2 + \sqrt{2}\, j\omega + 1}$$

$$H(j\omega) = \frac{1}{(1-\omega^2) + j\sqrt{2}\,\omega}$$

CÁLCULO DE MÓDULO: $|H(j\omega)| = \sqrt{Re^2 + Im^2}$

$$|H(j\omega)| = \frac{1}{\sqrt{(1-\omega^2)^2 + (\sqrt{2}\,\omega)^2}}$$

$$(1-\omega^2)^2 + (\sqrt{2}\,\omega)^2 = 1 - 2\omega^2 + \omega^4 + 2\omega^2$$
$$= 1 + \omega^4$$

$$|H(j\omega)| = \frac{1}{\sqrt{1+\omega^4}}$$

CÁLCULO DE FASE: $\measuredangle H = Tg^{-1}\left(\frac{Im(H)}{Re(H)}\right)$

$Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$ → DENOMINADOR

$0°$ POR SER Nº REAL → NUMERADOR

$$\measuredangle H = 0° - Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right) = -Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$$

ANÁLISIS DEL MÓDULO:

- BAJAS FRECUENCIAS: $\omega << 1$

$$|H(j\omega)| = \frac{1}{\sqrt{1+\omega^4}} = 1$$

$$= 20\log(1) = 0\,dB$$

- FRECUENCIA DE CORTE: $\omega = 1$

$$|H(j\omega)| = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \approx 0{,}707$$

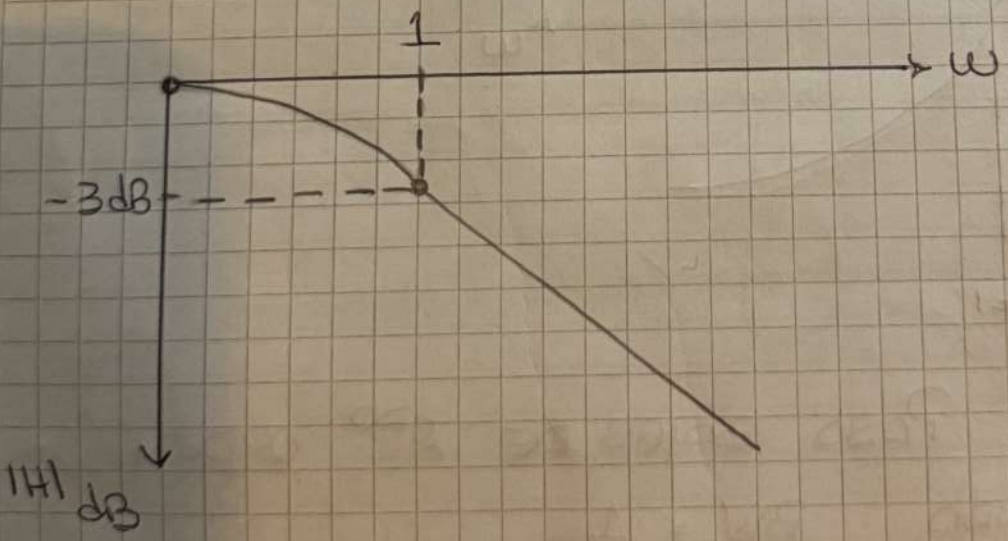
$$= 20\log(0{,}707) = -3dB$$

- ALTAS FRECUENCIAS: $\omega >> 1$

$$|H(j\omega)| = \frac{1}{\sqrt{\omega^4}} = \frac{1}{\omega^2} = \omega^{-2}$$

= PENDIENTE DECRECIENTE

$-40\,dB$.



ANÁLISIS DE FASE

- BAJAS FRECUENCIAS: $\omega \to 0$

$$\measuredangle H = -Tg^{-1}\left(\frac{0}{1}\right) = 0^\circ$$

- FRECUENCIA DE CORTE: $\omega = 1$

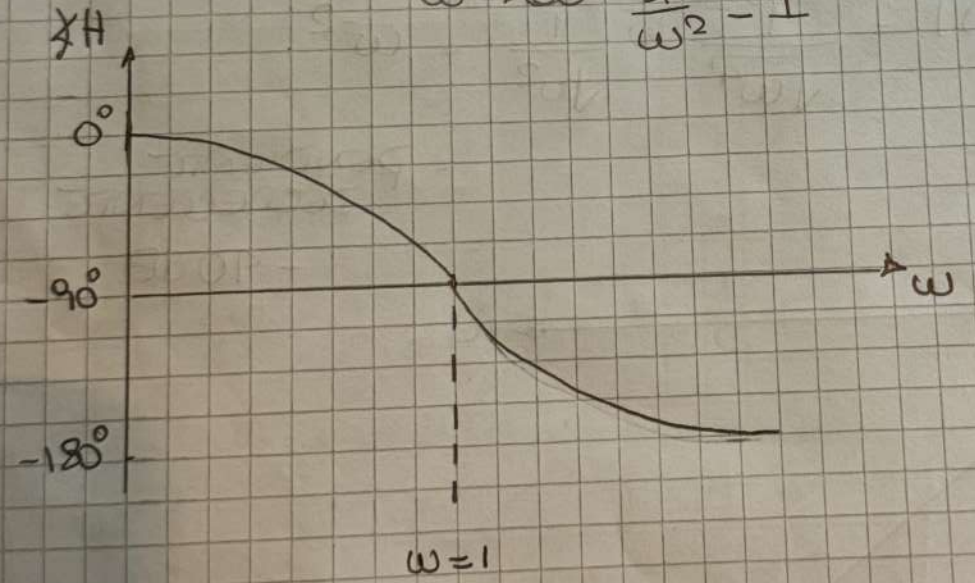
$$\measuredangle H = -Tg^{-1}\left(\frac{\sqrt{2}}{0}\right) = -90^\circ$$

$\frac{\sqrt{2}}{0} \to \infty$

- ALTAS FRECUENCIAS: $\omega \to \infty$

$$\measuredangle H = -Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right) = -180^\circ.$$

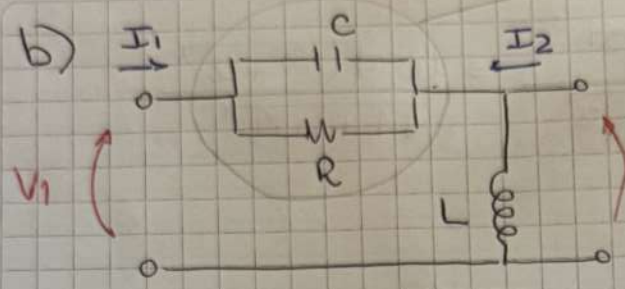
$$\lim_{\omega \to \infty} \frac{\sqrt{2}/\omega}{\frac{1}{\omega^2} - 1} = 0$$



TIPO DE FILTRO: PASA BAJOS DE 2DO ORDEN

ANCHO DE BANDA: $BW = 1$

b)



$$Z_{//}(s) = R \,//\, \frac{1}{sC} = \frac{R}{sCR+1}$$

$$H(s) = \frac{sL}{sL + \frac{R}{sCR+1}} = \frac{sL\,(sCR+1)}{s^2RLC + sL + R}$$

$$= \frac{s^2 + \frac{1}{RC}\cdot s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$R = 1\,\Omega$, $\omega_0 = 1$, $Q = \sqrt{2}/2$

- $\omega_0^2 = \frac{1}{LC} \Rightarrow LC = 1$.

- $\frac{\omega_0}{Q} = \sqrt{2} = \frac{1}{RC} = \frac{1}{C}$

$$\boxed{H(s) = \frac{s^2 + \sqrt{2}\,s}{s^2 + \sqrt{2}\,s + 1}}$$

EVALUANDO EN $s = j\omega$

$$H(j\omega) = \frac{-\omega^2 + j\sqrt{2}\,\omega}{(1-\omega^2) + j\sqrt{2}\,\omega}$$

CÁLCULO DE MÓDULO:

$$|H(j\omega)| = \frac{\sqrt{\omega^4 + 2\omega^2}}{\sqrt{(1-\omega^2)^2 + 2\omega^2}} = \frac{\omega\sqrt{\omega^2+2}}{\sqrt{1+\omega^4}}$$

CÁLCULO DE FASE:

NUMERADOR $\rightarrow$ $Tg^{-1}\left(\frac{\sqrt{2}\omega}{-\omega^2}\right) = Tg^{-1}\left(-\frac{\sqrt{2}}{\omega}\right)$

DENOMINADOR $\rightarrow$ $Tg^{-1}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$

$$\measuredangle H = \left(180^\circ - Tg^{-1}\left(\frac{\sqrt{2}}{\omega}\right)\right) - Tg^{-1}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$$

ANÁLISIS DEL MÓDULO:

- BAJAS FRECUENCIAS: $\omega \ll 1$

$$|H(j\omega)| = \omega\sqrt{2} \Rightarrow 20\log(\sqrt{2}) + 20\log(\omega)$$

$$\approx 3\,dB + 20\log(\omega)$$

PENDIENTE DE +20dB

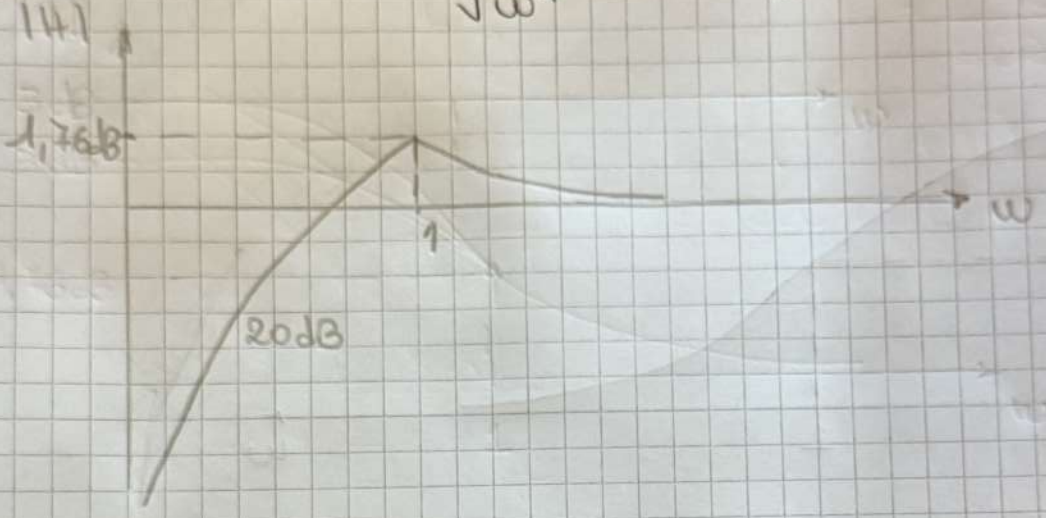
- FRECUENCIA DE CORTE: $\omega = 1$

$$|H(j\omega)| = \frac{1\sqrt{1+2}}{\sqrt{1+1}} = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}} = 1{,}225$$

$$= 1{,}76\,dB$$

- PARA ALTAS FRECUENCIAS: $\omega \gg 1$

$$|H(j\omega)| = \frac{\omega \cdot \sqrt{\omega^2}}{\sqrt{\omega^4}} = \frac{\omega^2}{\omega^2} = 1 = 0\,dB$$



PARA LA CLASIFICACIÓN OBSERVO LO QUE OCURRE EN LOS EXTREMOS DE FRECUENCIA.

ANÁLISIS DE FASE

- BAJAS FRECUENCIAS $\omega \to 0$

$$\measuredangle H = \left(180° - Tg^{-1}\left(\frac{\sqrt{2}}{0}\right)\right) - Tg^{-1}\left(\frac{\sqrt{2}\cdot 0}{1-0}\right)$$

$$= (180° - 90°) - 0° = 90°$$

- FRECUENCIA DE CORTE $\omega = 1$

$$\measuredangle H = \left(180° - Tg^{-1}\left(\sqrt{2}\right)\right) - Tg^{-1}\left(\frac{\sqrt{2}}{0}\right)$$

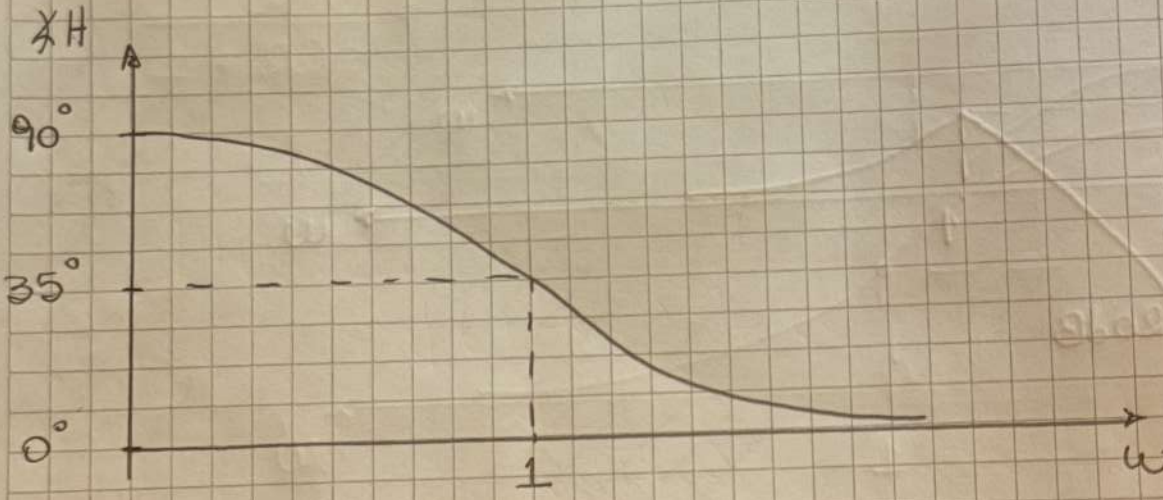
$$= (180° - 54,7°) - 90° = 35,3°$$

∞

- ALTAS FRECUENCIAS $\omega \to \infty$

$$\measuredangle H = \left(180^\circ - Tg^{-1}\left(\frac{\sqrt{2}}{\infty}\right)\right) - Tg^{-1}\left(\frac{\sqrt{2}\cdot\infty}{1-\infty^2}\right)$$

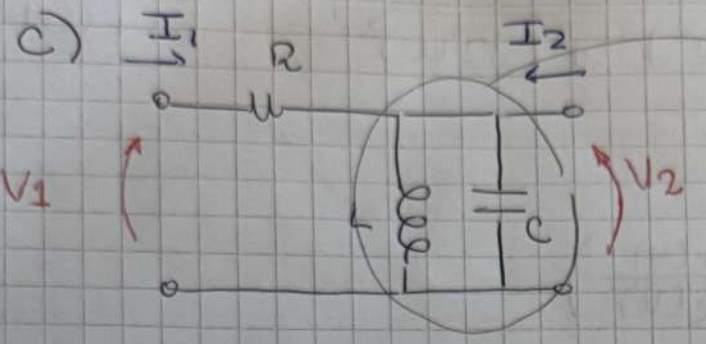
$$= (180^\circ - 0^\circ) - 180^\circ = 0^\circ$$



TIPO DE FILTRO: PASO ALTOS DE 2DO ORDEN

BW = ∞ → DESDE 1 HASTA ∞.

c)



$$Z_{//} = sL \,//\, \frac{1}{sC}$$

$$= \frac{sL}{s^2LC+1}$$

$$H(s) = \frac{\frac{sL}{s^2LC+1}}{R + \frac{sL}{s^2LC+1}} \cdot \frac{s^2LC+1}{s^2LC+1}$$

$$H(s) = \frac{sL}{R(s^2LC+1)+sL} = \frac{sL}{s^2RLC+sL+R}$$

$$H(s) = \frac{\frac{sL}{RLC}}{s^2 + s\frac{L}{RLC} + \frac{R}{RLC}} = \frac{s\,(1/RC)}{s^2 + s\,(1/RC) + 1/LC}$$

$$\omega_0^2 = \frac{1}{LC} = 1$$

$$\frac{\omega_0}{Q} = \frac{1}{RC} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$$

$$\boxed{H(s) = \frac{\sqrt{2}\,s}{s^2 + \sqrt{2}\,s + 1}}$$

EVALUANDO EN $s = j\omega$

$$H(j\omega) = \frac{j\sqrt{2}\,\omega}{-\omega^2 + \sqrt{2}\,j\omega + 1} = \frac{j\sqrt{2}\,\omega}{(1-\omega^2) + j\sqrt{2}\,\omega}$$

CÁLCULO DE MÓDULO

$$|H(j\omega)| = \frac{\sqrt{2}\,\omega}{\sqrt{(1-\omega^2)^2 + (\sqrt{2}\omega)^2}} = \frac{\sqrt{2}\,\omega}{\sqrt{1+\omega^4}}$$

CÁLCULO DE FASE

$$\measuredangle H = Tg^{-1}\left(\frac{Im(H)}{Re(H)}\right)$$

NUMERADOR → Nº IMAGINARIO PURO. (90°)

DENOMINADOR → $Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$

$$\measuredangle H = 90^\circ - Tg^{-1}\left(\frac{\sqrt{2}\,\omega}{1-\omega^2}\right)$$

Análisis del módulo

- Bajas frecuencias ($\omega \to 0$)

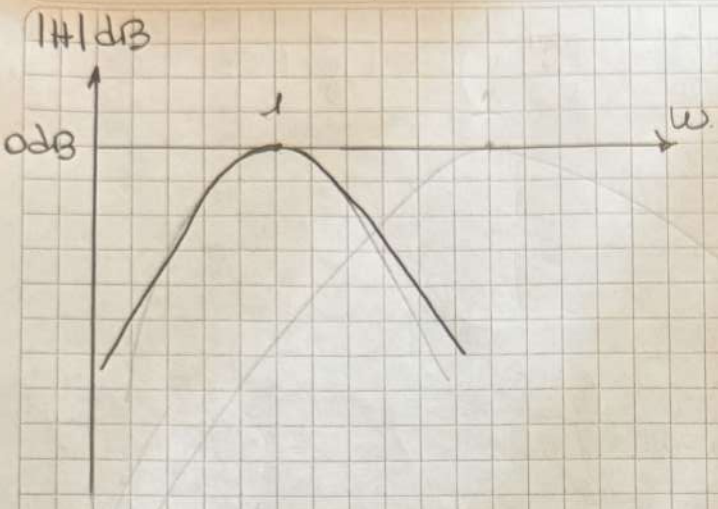
$$|H(j\omega)| = \frac{\sqrt{2}\cdot 0}{1} = 0 \to -\infty\ dB$$

- Frecuencia de corte $\omega = 1$

$$|H(j\omega)| = \frac{\sqrt{2}\cdot 1}{\sqrt{1+1}} = 1 \to 0\ dB$$

- Altas frecuencias $\omega \to \infty$

$$|H(j\omega)| = \frac{\sqrt{2}\,\omega}{\omega^2} = \frac{\sqrt{2}}{\omega} \to 0 \to -\infty\ dB$$



Análisis de fase

- Baja frecuencia $(\omega \to 0)$

$$\measuredangle H = 90^\circ - Tg^{-1}\left(\frac{\sqrt{2}\cdot 0}{1-0}\right) =$$

$$90^\circ - 0^\circ = 90^\circ$$

- Frecuencia de corte $\omega = 1$

$$\measuredangle H = 90^\circ - Tg^{-1}\left(\frac{\sqrt{2}}{1-1}\right) =$$

$$90^\circ - 90^\circ = 0^\circ$$

- Altas frecuencias $(\omega \to \infty)$

$$\measuredangle H = 90^\circ - Tg^{-1}\left(\frac{\infty}{1-\infty^2}\right) \to 2^{do} \text{ cuadrante}$$

$$= 90^\circ - 180^\circ = -90^\circ$$

- ALTAS FRECUENCIAS $\omega \to \infty$

$$\not\propto H = (180^\circ - \tan^{-1}\left(\frac{\sqrt{2}}{\infty}\right)) - \tan^{-1}($$

